

AUSTIN TONSAY

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EDUCATION:

California State Polytechnic University, Pomona

Expected Graduation: May 2029

Bachelor of Science, Aerospace Engineering (Emphasis: Astronautics) | GPA: 3.83

PERSONAL PROJECTS:

Baseline vs. Altitude-Optimized High-Power Rocketry Experiment:

May 2026 – Present

- Designing baseline and altitude-optimized high-power rockets predicted to reach 3,095 ft and 4,523 ft (~46% increase)
- Performing aerodynamic/stability trades across fin geometry, nose profile, body length, boattail, and surface finish in SolidWorks/OpenRocket.
- Iterating 5 custom motor-retention concepts and developing a bayonet-style retainer plus interchangeable 6-in modular avionics bay; manufacturing PETG-CF aerodynamic components with DFM and anisotropic loading.
- Calibrating simulations using component-level as-built mass/CG measurements and flight-tested baseline vehicle; converting vehicle loss after successful launch and apogee deployment into revised electronic tracking and high-visibility recovery requirements.

EXPERIENCE:

AIAA | Design, Build, Fly (DBF) Competition:

August 2025 – August 2026

Structures/Manufacturing Engineer

- Fabricated 3 material trade studies comparing 4–6 candidates across bending/torsional strength, mass, cost, and manufacturability to guide wing/fuselage structural material selection.
- Designed tail assembly in SolidWorks intended for 3D printing to reduce part count, enhance manufacturability, and improve aerodynamic performance from previous year.
- Owned wing/fuselage assembly, wood/foam core construction with carbon-fiber reinforcement, and control surface finishing with Monokote heat-shrink covering to reduce drag.

Bronco S.T.A.R. | BLADE Odyssey:

August 2025 – May 2026

Communications & Thermals Engineer

- Developed a heater trade study comparing 3 candidate heating solutions against a ~9.25 W / 37.2 Wh CubeSat power constraint, balancing thermal capability, power draw, size, and integration risk.
- Sized two 60 × 40 mm heating pads at 4.2 V, 3 W each (~0.71 A) within SWaP (size, weight, and power) constraints.
- Produced thermostat-control pseudocode to cycle battery heating between 0–10°C with a 30%-of-flight runtime failsafe for ~4-hour CubeSat mission.

Undergraduate Missiles, Ballistics, and Rocketry Association (UMBRA):

September 2025 – November 2025

Small Rocket Competition, Team Lead

- Pioneered model rocket fin and body geometries in OpenRocket, SolidWorks CAD, and Bambu Studio; improved projected apogee from 2,100 ft to 2,900 ft (~38% increase) through aerodynamic optimization.
- Led 5-person design/build team through aerodynamic design, manufacturing, and avionics integration while maintaining project schedule within 2 days of planned milestones.
- Programmed Arduino IDE for avionics systems to read rocket altitude, surrounding pressure, and temperature; post-flight failure analysis identified insufficient epoxy cure time as the cause of internal structural separation.

ACADEMIC PROJECTS:

ARO1021L - Analyzed OpenRocket, static-fire, inclinometer, and onboard altimeter apogee estimates across 3 model rockets; quantified method discrepancies and evaluated drag, manufacturing, and environmental sources of error.

EGR1000L - Drafted and programmed an Arduino-controlled animatronic system integrating 4 servos and an ~85% 3D-printed structure; iterated actuator selection and mechanical fit to achieve 3 coordinated motions within a 12-in design envelope.

SKILLS:

SolidWorks, OpenRocket, FDM Additive Manufacturing, Arduino IDE (C++), PrusaSlicer, Carbon Fiber Layups, Wood/Foam Airframe Fabrication, Excel VBA